Notes: Loewenstein & Qin (RFS, 2025)

An Equilibrium Model of Imperfect Hedging: Transaction Costs, Heterogeneity in Risk Aversion, and Return Volatility

Contents

1 Big picture

- **Question.** How do proportional transaction costs in derivative markets affect equilibrium trade, risk sharing, expected returns, Sharpe ratios, and return volatility?
- **Why this matters.** Policy proposals for financial transaction taxes often claim that higher transaction costs reduce short-term speculation and therefore reduce volatility. This paper shows that the opposite can occur in an equilibrium model of derivative trading.
- **Market.** Agents trade a zero-net-supply risky asset interpreted as a standardized exchange-traded derivative. The derivative has price process

  $$dS_t = \mu_0 S_t dt + \sigma_0 S_t dB_t,$$

  with transaction costs through ask price $(1+\theta)S_t$ and bid price $(1-\alpha)S_t$.
- **Hedging motive.** Agents have nontraded endowments, such as labor income or untraded underlying exposures, that are imperfectly correlated with the traded derivative. The derivative is useful but imperfect as a hedge.
- **Individual choice.** Without transaction costs, an agent continuously rebalances to maintain a constant dollar position. With transaction costs, continuous rebalancing is too costly, so the agent optimally trades only at buy and sell boundaries.
- **Key microeconomic object.** Each agent has a no-transaction region. The buy boundary, sell boundary, and frictionless target determine when the agent trades and how aggressively he hedges.
- **Equilibrium difficulty.** With zero net supply, the market must clear at both buy and sell boundaries. Transaction costs adjust expected returns differently at different boundaries, so clearing both boundaries is restrictive.
- **Main trade result.** Trade is sustained with positive transaction costs only when heterogeneity in risk aversion is sufficiently large. In the key equilibrium, the less risk-averse agent takes the short side.
- **Main volatility result.** When a trading equilibrium exists, the magnitude of equilibrium return volatility increases with transaction costs. Higher transaction costs require higher volatility to reconcile the Sharpe ratio implied by the outer and inner trading boundaries.
- **Sharpe ratio and liquidity premium.** The equilibrium Sharpe ratio changes with transaction costs through a boundary-width measure z. The liquidity premium is negative and compensates the short seller for providing liquidity and bearing rebalancing frictions.
- **No-trade equilibria.** With nontrivial transaction costs, equilibria without trade always exist. These equilibria can have a continuum of return volatility and Sharpe ratio values, but the paper focuses on when transaction costs still permit trade.
- **Policy takeaway.** Transaction costs do not mechanically stabilize derivative markets. In an equilibrium with imperfect hedging, higher transaction costs can raise volatility because volatility itself is part of how markets clear.

2 Detailed section-by-section guide

2.1 Introduction

- The paper begins from the policy debate over financial transaction taxes and the empirical ambiguity about whether transaction costs reduce or increase volatility.
- The authors emphasize that derivatives are central for risk sharing and are often subject to high transaction costs, yet most equilibrium asset-pricing models do not directly analyze proportional transaction costs in derivative markets.
- The paper proposes a tractable equilibrium model in which agents trade a standardized derivative to hedge nontraded endowments.
- The central intuition is that transaction costs create two market-clearing conditions: one at the buy side and one at the sell side. These conditions imply potentially different Sharpe ratios unless equilibrium volatility adjusts.
- The introduction previews the main result: when trade occurs, equilibrium return variance increases in transaction costs.

Interpretation: The introduction reframes transaction costs as an equilibrium pricing force rather than merely a trading friction. The key is that volatility is endogenous and helps markets clear.

2.2 Section 1: The model

- Section 1 first specifies the trading environment and the standardized derivative contract.
- Subsection 1.1 formulates the individual agent's problem with nontraded wealth, transaction costs, exponential utility, and a random horizon.
- Subsection 1.2 solves the frictionless benchmark. The agent holds a constant dollar amount in the risky asset, decomposed into a mean-variance component and a hedging component.
- Subsection 1.3 introduces transaction costs and shows that the optimal policy becomes a no-transaction region with buy and sell boundaries.
- Subsection 1.4 gives analytical boundary approximations. These are crucial because they allow equilibrium restrictions to be studied analytically.
- Subsection 1.5 discusses two implications of the individual choice problem: asymmetry between long and short positions, and how return volatility affects mean-variance and hedging demands.

Interpretation: Section 1 is the micro foundation. It shows why, with transaction costs, agents do not continuously rebalance but instead tolerate imperfect hedging inside a no-transaction region.

2.3 Section 2: Equilibrium

- Section 2 moves from the individual problem to a two-agent zero-net-supply equilibrium.
- Subsection 2.1 classifies equilibria with and without transaction costs and with and without trade. It also defines equilibrium as a pair (μ_0, σ_0) that clears the risky-asset market.
- Subsection 2.2 analyzes the equilibrium with transaction costs and trade. This is the main theoretical section.
- Proposition 3 verifies that aggregate terminal consumption is equal to aggregate endowment minus cumulative transaction costs, so there are no hidden resource flows in equilibrium.

- Proposition 4 gives a closed-form expression for the equilibrium return volatility. This expression is the paper's core analytical result.
- Proposition 5 studies the equilibrium Sharpe ratio and shows that the liquidity premium is negative.
- Subsection 2.3 uses numerical comparative statics to show how buy transaction costs, heterogeneity in risk aversion, endowment volatility, and revenue-maximizing transaction costs affect equilibrium outcomes.
- Subsection 2.4 analyzes equilibria without trade. Proposition 6 shows that such equilibria exist for any positive transaction cost.

Interpretation: Section 2 explains how transaction costs alter equilibrium itself. The model is not just partial-equilibrium portfolio choice: the derivative's return and volatility are chosen by market clearing.

2.4 Appendices

- Appendix A derives value functions with transaction costs for positive and negative trading boundaries.
- Appendix B provides the frictionless trading equilibrium, boundary approximation diagnostics, and further analysis of endogenous return volatility.
- Appendix C proves the propositions used in the main text.
- Appendix figures A1 and B1 validate the approximations used to obtain Proposition 4 and compare analytical formulas with numerical equilibrium solutions.

Interpretation: The appendices are not peripheral; they justify the approximations that make the equilibrium analytically tractable.

3 Model primitives and measurement

3.1 Assets and contract standardization

- There is a riskless asset with zero interest rate.
- There is a risky asset in zero net supply, interpreted as an exchange-traded standardized derivative.
- The derivative has price dynamics
$$dS_t = \mu_0 S_t dt + \sigma_0 S_t dB_t. \tag{1}$$
- The return volatility σ_0 can be positive or negative. Economically, its sign captures the direction of risk exposure relative to the Brownian shock.
- The contract's expiration date equals the agents' horizon for simplicity.
- Standardization means the contract is not customized to perfectly hedge every agent's endowment. This imperfectness is what creates dynamic trading.

Interpretation: The derivative is useful because it is correlated with endowment risk, but it is not a perfect hedge. This makes rebalancing relevant.

3.2 Transaction costs

The market posts ask and bid prices:

$$S_t^A = (1+\theta)S_t, \qquad S_t^B = (1-\alpha)S_t, \tag{2}$$

where $\theta \geq 0$ is the proportional cost of buying and $0 \leq \alpha < 1$ is the proportional cost of selling.

Interpretation: The total round-trip transaction cost is governed by $\theta + \alpha$. When these costs rise, agents rebalance less frequently, but equilibrium volatility must also adjust for the market to clear with trade.

3.3 Nontraded wealth and imperfect hedging

Each agent has nontraded wealth:
$$dY_t = \mu dt + \sigma d\widetilde{B}_t. \tag{3}$$
The risky asset Brownian motion is correlated with the nontraded endowment:

$$dB_t = \rho d\widetilde{B}_t + \sqrt{1-\rho^2}\, dB_t^o. \tag{4}$$

Interpretation: The parameter $\rho\sigma\sigma_0$ determines whether the risky asset hedges or amplifies endowment risk. If the traded derivative is negatively correlated with nontraded wealth, it has hedging value.

3.4 Portfolio evolution with transaction costs

Let x_t be dollars in the riskless asset and y_t be dollars invested in the risky asset. Let I_t and D_t be cumulative purchases and sales. Then

$$dx_t = (1-\alpha)dD_t - (1+\theta)dI_t, \tag{5}$$
$$dy_t = \mu_0 y_t dt + \sigma_0 y_t dB_t + dI_t - dD_t. \tag{6}$$

Interpretation: Purchases reduce the riskless account by the ask price; sales increase the riskless account by the bid price. The state variable y_t evolves both because of price movements and because of trading.

3.5 Utility, random horizon, and adjusted expected return

Agents have negative exponential utility with absolute risk aversion A. The terminal date τ is exponentially distributed with parameter λ. After a change of measure that absorbs nontraded wealth, the risky-asset drift becomes

$$\bar{\mu} \equiv \mu_0 - \rho A \sigma \sigma_0. \tag{7}$$

Interpretation: The adjusted expected return $\bar{\mu}$ combines speculative compensation μ_0 and hedging demand from nontraded wealth. This is the key state variable for individual portfolio choice.

function of an arithmetic Brownian motion. To model the need of imperfect hedging, Subsection 1.1 will assume each agent has an endowment that follows an arithmetic Brownian motion correlated with the Brownian motion B_t. As a result, this setting enables us to study how the agents optimally trade off dynamically between transaction costs and hedge quality. Further, these assumptions lead to a fairly simple contract description that seems to be an important feature of standardized contracts.

The contract specification in Equation (1) implies strictly positive prices as for an exchange-traded option, opposed to the value of an uncollateralized forward contract.[17] This feature is important when examining the impact of transaction costs proportional to the dollar amount transacted, since it ensures positive transaction costs and a well-defined liquidity premium. Specifically we assume the exchange continuously posts ask prices $S_t^A = (1+\theta)S_t$ and bid prices $S_t^B(1-\alpha)S_t$, where $\theta \geq 0$ and $0 \leq \alpha < 1$ represent proportional transaction cost rates.

We assume that the derivative pays no dividends.[18] We take its expected return μ_0 and return volatility σ_0 to define the derivative contract S. The return volatility σ_0 can be thought of as proportional risk exposure, and the coefficient μ_0 is a risk premium. The initial price S_0 captures the contract size and is assumed to be strictly positive; since transaction costs are proportional to dollars transacted, the initial price does not play an important role in our analysis.[19]

While one might design custom-tailored contracts (for example, with stochastic expected return and return volatility), these contracts might be more appropriate for an over-the-counter market. As observed in Duffie and Huang

4 Individual portfolio problem

4.1 Frictionless benchmark

Without transaction costs, the HJB equation is

$$-\lambda u - \lambda \frac{1}{A} e^{-Aw} + \max_{y} \left\{ \bar{\mu} y u_w + \frac{1}{2} \sigma_0^2 y^2 u_{ww} \right\} = 0. \tag{8}$$

Proposition 1 gives the optimal dollar investment:

$$y^* = \frac{\bar{\mu}}{A\sigma_0^2} = \frac{\mu_0}{A\sigma_0^2} - \frac{\rho\sigma}{\sigma_0}. \tag{9}$$

Interpretation: The position has two parts. The first is mean-variance demand; it depends on expected return and risk aversion. The second is hedging demand; it depends on the correlation between the traded derivative and nontraded wealth.

The associated Hamilton-Jacobi-Bellman (HJB) equation turns out to be

$$-\lambda u - \lambda \frac{1}{A} e^{-Aw} + \max_{\substack{x,y \\ s.t\ x+y=w}} \left(\bar{\mu} y u_w + \frac{1}{2} \sigma_0^2 y^2 u_{ww} \right) = 0. \tag{8}$$

Proposition 1 presents the analytical solution to Equation (8).

Proposition 1. When there are no transaction costs—that is, $\theta = \alpha = 0$, the optimal wealth invested in the risky asset is given as a function of the adjusted rate of expected return and the risk aversion:

$$y^* = \frac{\bar{\mu}}{A\sigma_0^2} = \frac{\mu_0}{A\sigma_0^2} - \frac{\rho\sigma}{\sigma_0}. \tag{9}$$

Moreover, the value function is given as

$$u = -\frac{2\lambda\sigma_0^2}{A\left(2\lambda\sigma_0^2 + \bar{\mu}^2\right)} e^{-Aw} = -\frac{2\lambda}{A\left(2\lambda + (\mu_0/\sigma_0 - \rho A\sigma)^2\right)} e^{-Aw}. \tag{10}$$

4.2 Why transaction costs create a no-transaction region

- In the frictionless benchmark, the agent continuously rebalances to maintain y^*.
- With proportional costs, continuous rebalancing would generate infinite transaction costs.
- The agent instead allows the risky position to drift within a no-transaction band.
- If the position falls below the buy boundary y_b, the agent buys up to y_b.
- If the position rises above the sell boundary y_s, the agent sells down to y_s.
- Between y_b and y_s, the agent accepts imperfect hedging to avoid paying costs.

Interpretation: Transaction costs convert portfolio choice from a static target problem into a dynamic boundary problem.

4.3 Analytical boundary approximations

Proposition 2 gives analytical optimal buy and sell boundaries up to constants C_0 and C_3. For a long agent, the buy and sell boundaries depend on $\bar{\mu}$, A, σ_0, λ, and the transaction-cost rates θ and α. A corresponding expression applies to a short seller.

- The outer boundaries, long-sell and short-buy, are exact.
- The inner boundaries, long-buy and short-sell, are approximations.
- The approximations are accurate for reasonable parameter values and are validated in Appendix Figure A1.
- The boundary formulas imply conditions under which the inner boundaries are nonzero, which becomes important in the no-trade equilibrium.

Interpretation: The boundary formulas are the bridge from individual optimization to equilibrium asset pricing. Market clearing will set one agent's boundary equal to the negative of the other agent's boundary.

Proposition 2. If an agent goes long, the analytical optimal buy and sell boundaries up to two constants to be determined (C_0 and C_3) are

$$y_b \approx \frac{\bar{\mu}+\frac{\lambda}{AC_0}\frac{\theta+\alpha}{1+\theta}-\sqrt{\left(\bar{\mu}+\frac{\lambda}{AC_0}\frac{\theta+\alpha}{1+\theta}\right)^2+2\lambda\sigma_0^2\frac{AC_0+1}{AC_0}}}{\sigma_0^2 A(1+\theta)} \text{ and } y_s = \frac{\bar{\mu}+\sqrt{\bar{\mu}^2+2\lambda\sigma_0^2\frac{AC_3+1}{AC_3}}}{\sigma_0^2 A(1-\alpha)}. \tag{15}$$

If an agent short sells, the optimal buy and sell boundaries up to two constants (C_0 and C_3) are

$$y_b = \frac{\bar{\mu}-\sqrt{\bar{\mu}^2+2\lambda\sigma_0^2\frac{AC_0+1}{AC_0}}}{\sigma_0^2 A(1+\theta)} \text{ and } y_s \approx \frac{\bar{\mu}-\frac{\lambda}{AC_3}\frac{\theta+\alpha}{1-\alpha}+\sqrt{\left(\bar{\mu}-\frac{\lambda}{AC_3}\frac{\theta+\alpha}{1-\alpha}\right)^2+2\lambda\sigma_0^2\frac{AC_3+1}{AC_3}}}{\sigma_0^2 A(1-\alpha)}. \tag{16}$$

Moreover, the bounds on the constants are given by

$$-\frac{2\lambda\sigma_0^2}{A(2\lambda\sigma_0^2+\bar{\mu}^2)} \geq C_j \geq -\frac{1}{A}, \text{ for } j=0,3. \tag{17}$$

For proof, see Subsection C.1 in the Appendix. The constants C_0 and C_3 are introduced in Section A of the Appendix for both cases when an agent goes long or short.

Comparing the expressions for the outer (long-sell and short-buy) boundaries to those for the inner (long-buy and short-sell) boundaries, which are closer to the zero position, one can see that the long-buy (short-sell) boundary has a similar form as the long-sell (short-buy) boundary, except that the adjusted expected return $\bar{\mu}$ is further adjusted by the term $\frac{\lambda}{\cdot}\frac{\theta+\alpha}{\cdot} <$

4.4 Long-short asymmetry

- With transaction costs, long and short positions are not symmetric.

- Short positions tend to have narrower no-transaction regions when risk aversion is held fixed.
- If two agents have identical risk aversion but opposite hedging motives, the market need not clear with zero expected return.
- A trading equilibrium therefore typically requires heterogeneity in risk aversion.

Interpretation: The asymmetry between long and short positions is what makes risk-aversion heterogeneity central. The less risk-averse agent can take the short side and provide liquidity.

5 Equilibrium setup

5.1 Two agents and zero net supply

The economy has two agents, indexed by L and S. They may differ in risk aversion A_i, endowment volatility σ_i, and correlation ρ_i. The risky derivative is in zero net supply.

- In a trading equilibrium, one agent is long and the other is short.
- Market clearing requires that the dollar positions sum to zero at relevant boundaries.
- The equilibrium objects are the derivative expected return μ_0 and return volatility σ_0.

Interpretation: The derivative contract is endogenous in equilibrium: its expected return and volatility must support optimal trades that clear the market.

5.2 Definition of equilibrium

Definition 1 says an equilibrium is a pair (μ_0, σ_0) such that agents optimize and markets clear. The risky asset market clears because one agent's position is the negative of the other's position.

characteristics or any equilibrium quantities. A formal definition of equilibrium is given as follows.

Definition 1. In the economy with two agents, an equilibrium is a set of expected return and return volatility for the risky asset (μ_0, σ_0) such that: (i)

Interpretation: The riskless asset does not appear directly in the definition because aggregate consumption equals aggregate endowment minus transaction costs, with transaction costs consumed by an institutional agent.

5.3 Equilibrium classes

The paper classifies equilibria into four conceptual cases:

1. no transaction costs and no trade;
2. no transaction costs and trade;
3. transaction costs and trade;
4. transaction costs and no trade.

Interpretation: The economically central case is transaction costs with trade. But the no-trade case is important because it shows that transaction costs can shut down risk-sharing completely.

6 Equilibrium with transaction costs and trade

6.1 Market clearing at outer and inner boundaries

The outer-boundary clearing condition yields an expression for the equilibrium Sharpe ratio:

$$\frac{\mu_0}{\sigma_0} = \frac{A_L A_S(\rho_L \sigma_L z + \rho_S \sigma_S)}{A_L + A_S z}. \tag{10}$$

The boundary-width measure is

$$z = \frac{1 + w_2^L}{1 + w_2^S} = -\frac{A_L \bar{\mu}_S}{A_S \bar{\mu}_L}. \tag{11}$$

The inner-boundary condition yields a second Sharpe-ratio expression, adjusted for transaction costs through η_L and η_S.

Rearranging the market clearing condition at the outer boundaries (23) yields an expression for the Sharpe ratio:

$$\frac{\mu_0}{\sigma_0}=\frac{A_L A_S(\rho_L\sigma_L z+\rho_S\sigma_S)}{A_L+A_S z}, \text{ where } z\equiv\frac{1+w_2^L}{1+w_2^S}=\frac{\frac{y_s^L}{\frac{\bar{\mu}_L}{A_L\sigma_0^2}}}{\frac{y_b^S}{\frac{\bar{\mu}_S}{A_S\sigma_0^2}}}=-\frac{y^{*S}}{y^{*L}}=-\frac{A_L\bar{\mu}_S}{A_S\bar{\mu}_L}. \tag{27}$$

Note that z measures the width of the trading boundaries and is an important equilibrium quantity that characterizes the equilibrium Sharpe ratio, return volatility, and liquidity premium. The market clearing condition at the outer boundaries, $y_s^L=-y_b^S$, simplifies the expression for z. Moreover, the market clearing condition at the inner boundaries, $y_b^L=-y_s^S$, yields $z=\frac{1-w_1^L}{1-w_1^S}$.

In light of Proposition 2, it is useful to rewrite the market clearing condition at the inner boundaries by adjusting the expected return for round-trip transaction costs with $\eta_L\approx\frac{\lambda(\theta+\alpha)}{-A_L C_{0,L}(1+\theta)}>0$ and $\eta_S\approx\frac{\lambda(\theta+\alpha)}{-A_S C_{3,S}(1-\alpha)}>0$ as (see also Subsection B.2 in the Appendix for more details)

$$\frac{\mu_0-\eta_L}{A_L\sigma_0^2}(1-\hat{w}_1^L)+\frac{\mu_0+\eta_S}{A_S\sigma_0^2}(1-\hat{w}_1^S)=\frac{\rho_L\sigma_L}{\sigma_0}(1-\hat{w}_1^L)+\frac{\rho_S\sigma_S}{\sigma_0}(1-\hat{w}_1^S), \tag{28}$$

where the round-trip transaction costs adjusted weights satisfy $1-\hat{w}_1^L=\frac{y_b^L}{\frac{\bar{\mu}_L-\eta_L}{A_L\sigma_0^2}}$ and $1-\hat{w}_1^S=\frac{y_s^S}{\frac{\bar{\mu}_S+\eta_S}{A_S\sigma_0^2}}$. Corollary 1 shows that the inequalities $\bar{\mu}_S+\eta_S<0$ and $\bar{\mu}_L-\eta_L>0$ imply nonzero trading boundaries. As discussed in Subsection 2.1, the weight w_2^i captures the fractional deviation of the outer boundary from the optimal frictionless portfolio. Likewise, the adjusted weight $\hat{w}_1^i$ captures the fractional deviation of the inner boundary from the optimal frictionless portfolio when the expected return is adjusted for round-trip transaction costs. Inspection reveals $\hat{w}_1^L<w_1^L$ and $\hat{w}_1^S<w_1^S$, suggesting that Equation (28) implies another expression for the Sharpe ratio:

$$\frac{\mu_0}{\sigma_0}=\frac{A_L A_S(\rho_L\sigma_L\hat{z}+\rho_S\sigma_S)}{A_S\hat{z}+A_L} \\ +\frac{1}{\sigma_0}\frac{A_S\eta_L\hat{z}-A_L\eta_S}{A_S\hat{z}+A_L}, \text{ where } \hat{z}\equiv\frac{1-\hat{w}_1^L}{1-\hat{w}_1^S}=\frac{-A_L(\bar{\mu}_S+\eta_S)}{A_S(\bar{\mu}_L-\eta_L)}. \tag{29}$$

Note that $\hat{z}$ is also an important equilibrium quantity that measures the

Interpretation: Without transaction costs, one Sharpe ratio expression is enough. With transaction costs, inner and outer boundaries imply different adjusted Sharpe ratios. Equilibrium must reconcile them.

6.2 Proposition 4: endogenous return volatility

Proposition 4 equates the two Sharpe ratio expressions and solves for equilibrium return volatility:

$$\sigma_0 = \text{function of } (\theta, \alpha, A_L, A_S, \rho_L, \rho_S, \sigma_L, \sigma_S, z, \hat{z}, C_{0,L}, C_{3,S}). \tag{12}$$

- Transaction-cost adjustments enter through η_L and η_S.
- The expression depends on the gap between z and $\hat{z}$, the unadjusted and transaction-cost-adjusted boundary-width measures.
- Higher transaction costs increase the wedge between the Sharpe ratios implied by the two boundary-clearing conditions.
- Equilibrium return volatility rises in magnitude to offset that wedge and preserve market clearing.

Interpretation: This is the paper's core theorem. Volatility is endogenous because it changes both hedging and mean-variance demands. Higher volatility can be required to make both sides of the derivative market clear.

1. In an equilibrium with transaction costs and trade, the Sharpe ratio and return volatility are determined, and the magnitude of equilibrium return volatility increases in transaction costs. When there are transaction costs, due to the asymmetry between the long and short positions, the less risk-averse agent must be short.[30] In this case, neither of the two market clearing conditions is redundant, in contrast to the equilibrium without transaction costs and with trade. In equilibrium, the two Sharpe ratio expressions implied by both the inner and outer boundaries must be equal—that is,

$$\underbrace{\frac{A_L A_S(\rho_L\sigma_L z+\rho_S\sigma_S)}{A_S z+A_L}}_{\text{Sharpe ratio implied by outer boundaries}} = \underbrace{\frac{A_L A_S(\rho_L\sigma_L\hat{z}+\rho_S\sigma_S)}{A_S\hat{z}+A_L}+\frac{1}{\sigma_0}\frac{A_S\eta_L\hat{z}-A_L\eta_S}{A_S\hat{z}+A_L}}_{\text{Sharpe ratio implied by inner boundaries}}. \tag{30}$$

Rearranging Equation (30) yields Proposition 4, which presents an analytical expression for the equilibrium return volatility based on the accurate approximation of trading boundaries, where the role of the round-trip transaction cost adjustments at the inner boundaries is prominent.

Proposition 4. In the equilibrium with transaction costs and trade, equating the Sharpe ratio expressions implied by the outer and the approximate inner boundaries implies that the equilibrium return volatility can be analytically expressed as a function of transaction costs when $z \neq \hat{z}$:

$$\sigma_0 = \frac{\frac{A_S\lambda(\theta+\alpha)\hat{z}}{-A_L C_{0,L}(1+\theta)} - \frac{A_L\lambda(\theta+\alpha)}{-A_S C_{3,S}(1-\alpha)}}{\left[\frac{A_L A_S(\rho_L\sigma_L z+\rho_S\sigma_S)}{A_L+A_S z} - \frac{A_L A_S(\rho_L\sigma_L\hat{z}+\rho_S\sigma_S)}{A_L+A_S\hat{z}}\right](A_L+A_S\hat{z})}. \tag{31}$$

Equation (31) makes clear that the return volatility is determined by transaction costs and differences in risk aversion. As transaction costs increase, the two Sharpe ratios implied by different boundaries differ unless the magnitude of return volatility adjusts to offset the effect of increased transaction costs.Intuitively, one can think of the adjusted expected return and, thus, the effective mean-variance demands at the inner boundaries relative to those at the outer boundaries as being further adjusted for round-trip transaction costs. Since mean-variance demands move in 1/variance while hedging demands move in 1/volatility,[31] to match the Sharpe ratio expressions at the two

6.3 Proposition 5: Sharpe ratio and liquidity premium

Proposition 5 provides bounds on the equilibrium Sharpe ratio and shows that the liquidity premium is negative. The Sharpe ratio decreases or increases in z depending on the sign of

$$\rho_L A_L \sigma_L - \rho_S A_S \sigma_S. \tag{13}$$

- The liquidity premium compares the equilibrium Sharpe ratio with the no-transaction-cost Sharpe ratio.
- It is negative and compensates the short seller.

- The short seller is providing liquidity because the less risk-averse agent typically absorbs the short position needed for market clearing.

Interpretation: The model endogenizes a liquidity premium from transaction costs and imperfect hedging. It is not imposed as an exogenous spread.

3. In an equilibrium with trade, Sharpe ratio decreases (increases) in transaction costs when $\rho_L A_L \sigma_L < (>) \rho_S A_S \sigma_S$; the liquidity premium is always negative and compensates the short agent.In an equilibrium with transaction costs and trade, the measure of trading boundary width z increases in transaction costs and $1 < z < \frac{1+\theta}{1-\alpha}$ (see Subsection B.2 and the left panel of Figure B.1 in the Appendix for more details). Given the expression for equilibrium Sharpe ratio as a function of the measure z (Equation (27)) and the bounds on z, Proposition 5 provides bounds on the equilibrium Sharpe ratio and shows the liquidity premium is negative.

Proposition 5. In an equilibrium with transaction costs and trade, the equilibrium Sharpe ratio decreases (increases) in the measure z when $\rho_L A_L \sigma_L < (>) \rho_S A_S \sigma_S$. Given that the measure z is monotonic in transaction costs, then

$$\frac{\mu_0^{NTC}}{\sigma_0^{NTC}} > (<) \frac{\mu_0}{\sigma_0} > (<) \frac{A_L A_S (\rho_L \sigma_L \frac{1+\theta}{1-\alpha} + \rho_S \sigma_S)}{A_L + A_S \frac{1+\theta}{1-\alpha}}, \text{ when } \rho_L A_L \sigma_L < (>) \rho_S A_S \sigma_S. \tag{32}$$

The equilibrium liquidity premium

$$\sigma_0 \left(\frac{A_L A_S (\rho_L \sigma_L z + \rho_S \sigma_S)}{A_L + A_S z} - \frac{A_L A_S (\rho_L \sigma_L + \rho_S \sigma_S)}{A_L + A_S} \right) = \mu_0 - \sigma_0 \frac{\mu_0^{NTC}}{\sigma_0^{NTC}}$$

is negative and compensates the short seller.

6.4 Why risk-aversion heterogeneity is necessary

- The model's leading trading equilibrium requires $A_S < A_L$: the short seller is less risk averse.
- If risk aversions are too similar, the widths of the agents' no-transaction regions cannot be matched while clearing the market.
- Near the no-trade region, equilibrium volatility can explode or the trading equilibrium ceases to exist.

Interpretation: Heterogeneity is not a cosmetic assumption. It is required to generate compatible trading motives in the presence of transaction costs.

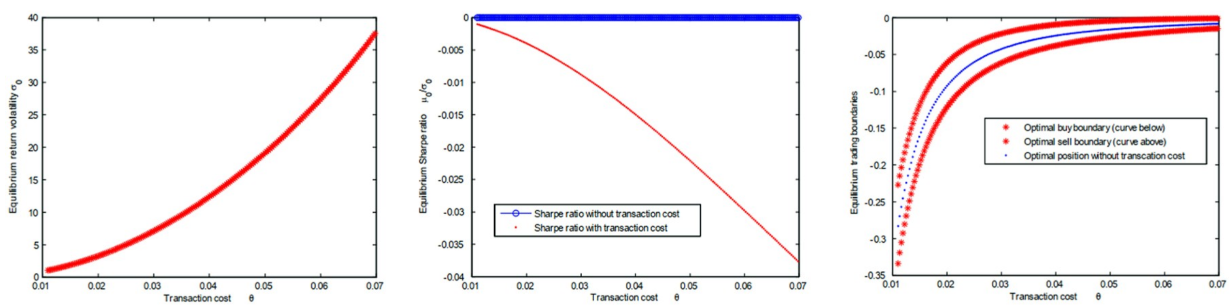
7 Comparative statics for trading equilibria

7.1 Impact of buy transaction cost

Read: Figure 2 studies how increasing the buy transaction cost θ affects equilibrium outcomes.

- The left panel shows that equilibrium return volatility increases with θ.
- The middle panel shows that the equilibrium Sharpe ratio changes with transaction costs through the boundary-width measure.
- The right panel shows that trading boundaries adjust as transaction costs widen the no-transaction region.

Economic message: Higher transaction costs do not stabilize the market in this equilibrium. Instead, higher volatility is needed to support hedging demand and reconcile the two market-clearing conditions.



7.2 Impact of risk aversion heterogeneity

Read: Figures 3 and 4 vary one agent's risk aversion while holding the other agent's risk aversion fixed.

- Figure 3 shows the trading boundaries, equilibrium volatility, and Sharpe ratio as heterogeneity changes.
- The less risk-averse agent tends to be the short seller.
- As risk aversions become similar, the trading equilibrium approaches the no-trade region.
- Figure 4 decomposes transaction costs into initial, intermediate, and terminal components and plots utility bounds.

Economic message: Risk aversion heterogeneity creates the motive for trade. When agents are too similar, transaction costs eliminate the gains from trade.

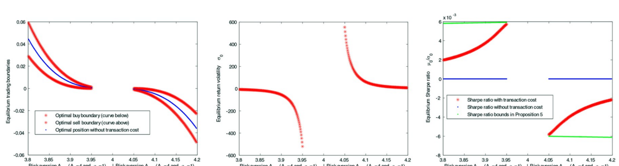


Figure 3: risk-aversion comparative statics, trading boundaries and asset prices.

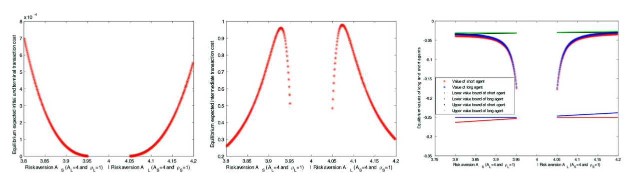


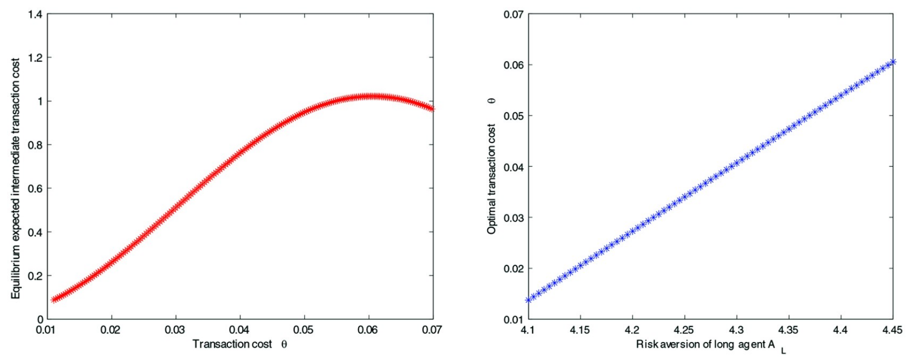
Figure 4: risk-aversion comparative statics, costs and utility.

7.3 Revenue-maximizing transaction cost

Read: Figure 5 asks what buy transaction cost maximizes expected transaction-cost revenue for the institutional agent.

- The expected intermediate transaction cost is hump-shaped in θ.
- Very low transaction costs generate little revenue per trade.
- Very high transaction costs reduce trade too much.
- The revenue-maximizing transaction cost rises when the difference in risk aversion rises because agents have stronger incentives to trade.

Economic message: An exchange or tax authority may maximize revenue at an interior transaction-cost level. This level is not necessarily welfare-maximizing and can be associated with higher volatility.



7.4 Impact of endowment volatility

- Higher endowment volatility increases the value of hedging.
- Stronger hedging motives make trade easier to sustain despite transaction costs.
- The Internet Appendix studies additional comparative statics for correlations and endowment volatility.

Interpretation: Endowment risk is the source of hedging demand. More endowment risk makes the derivative market more useful, but transaction costs still distort the equilibrium.

8 Equilibria without trade

8.1 Existence of no-trade equilibria

Proposition 6 shows that with nontrivial transaction costs, equilibria without trade always exist.

- Agents choose not to trade when the equilibrium return volatility and expected return make crossing the transaction-cost boundaries unattractive.
- In no-trade equilibria, neither agent takes a long or short position.
- There can be a continuum of no-trade equilibria because many values of σ_0 and μ_0 can be consistent with no agent wanting to trade.

Interpretation: Transaction costs can support markets that exist in name but do not facilitate risk sharing. This is important for benchmark selection: comparing an equilibrium with costs to a frictionless equilibrium may miss no-trade outcomes.

therefore, holding the initial endowment is optimal.

Proposition 6. In a model with nontrivial transaction costs, that is, $\alpha+\theta>0$, there always exist equilibria without trade. Assign one agent to be S and the other to be L. If $\rho_L A_L \sigma_L < \rho_S A_S \sigma_S$, choose γ so that $\rho_L A_L \sigma_L < \gamma < \rho_S A_S \sigma_S$ and the equilibrium return volatility $\sigma_0 > 0$ to satisfy

$$\sigma_0 \le \max\left(\frac{(\alpha+\theta)\lambda}{(1+\theta)(\gamma-\rho_L A_L \sigma_L)}, -\frac{(\alpha+\theta)\lambda}{(1-\alpha)(\gamma-\rho_S A_S \sigma_S)}\right). \qquad (34)$$

If $\rho_S A_S \sigma_S < \rho_L A_L \sigma_L$, choose γ so that $\rho_S A_S \sigma_S < \gamma < \rho_L A_L \sigma_L$ and the equilibrium return volatility $\sigma_0 < 0$ to satisfy

$$\sigma_0 \ge \max\left(\frac{(\alpha+\theta)\lambda}{(1+\theta)(\gamma-\rho_L A_L \sigma_L)}, -\frac{(\alpha+\theta)\lambda}{(1-\alpha)(\gamma-\rho_S A_S \sigma_S)}\right), \qquad (35)$$

and set $\mu_0 = \gamma\sigma_0$, then the agents do not trade in the risky asset with an expected return μ_0 and a return volatility σ_0. Further, if $\rho_S A_S \sigma_S = \rho_L A_L \sigma_L$, there are no equilibria with trade.

Proof. Similar to the proof of inequality (33) in Subsection 2.2 Item 5, these results follow from the conditions for the long-buy and short-sell boundaries to be zero in Corollary 1: $\frac{\lambda(\theta+\alpha)}{1+\theta} \ge \bar{\mu}_L = \mu_0 - \rho_L A_L \sigma_L \sigma_0 \ge 0$ and $-\frac{\lambda(\theta+\alpha)}{1-\alpha} \le \bar{\mu}_S = \mu_0 - \rho_S A_S \sigma_S \sigma_0 \le 0$. These inequalities can be rearranged to find values of the equilibrium expected return μ_0 and the equilibrium return volatility σ_0, which are consistent with these conditions. ■

Proposition 6 shows that equilibria with transaction costs and without trade

8.2 Economic interpretation

- In the trading equilibrium, volatility is large enough to make hedging and mean-variance demand sufficiently strong to generate trade.
- In the no-trade equilibrium, volatility is too small or expected returns are not attractive enough to overcome transaction costs.
- As transaction costs converge to zero, a sequence of no-trade equilibria can converge to a zero-return, zero-volatility benchmark.

Interpretation: The model contains both active and inactive markets. Which equilibrium is relevant depends on the contract specification and the economic environment.

9 Appendix results and analytical validation

9.1 Frictionless equilibrium with trade

Proposition 7 characterizes the no-transaction-cost equilibrium with trade. It gives a constant Sharpe ratio and each agent's equilibrium demand.

B.1 Equilibrium without transaction costs and with trade

We characterize the equilibrium without transaction costs and with trade mentioned in Subsection 2.1 in Proposition 7.

Proposition 7. In the equilibrium without transaction costs and with trade, the Sharpe ratio is given as a constant in Equation (25). Agent i's demand for the risky asset in equilibrium is given in Equation (26). Moreover, agent i's equilibrium expected utility is given by

$$u_i^{NTC} = -\frac{2\lambda\left(\sigma_0^{NTC}\right)^2}{A_i\left(2\lambda\left(\sigma_0^{NTC}\right)^2+\left(\bar{\mu}_i^{NTC}\right)^2\right)} = -\frac{2\lambda\left(\sigma_0^{NTC}\right)^2}{A_i\left(2\lambda\left(\sigma_0^{NTC}\right)^2+\left(y_i^{NTC}A_i\left(\sigma_0^{NTC}\right)^2\right)^2\right)},\ \text{for } i=L,S,$$

where the adjusted expected return for each agent is given as

$$\bar{\mu}_i^{NTC} = \frac{\left(A_j\rho_j\sigma_j - A_i\rho_i\sigma_i\right)A_i\sigma_0^{NTC}}{A_S+A_L},\ \text{for } i=L,S,\ \text{and } j=S,L.$$

Proof. When there are no transaction costs, Proposition 1 gives each agent's demand:

$$y_i^{NTC} = \frac{\bar{\mu}_i^{NTC}}{A_i\left(\sigma_0^{NTC}\right)^2} = \frac{\mu_0^{NTC}}{A_i\left(\sigma_0^{NTC}\right)^2} - \rho_i\frac{\sigma_i}{\left(\sigma_0^{NTC}\right)^2},\ \text{for } i=L,S.$$

Adding the demands across the agents, by the market clearing condition, we have

$$\frac{\mu_0^{NTC}}{A_L\left(\sigma_0^{NTC}\right)^2} + \frac{\mu_0^{NTC}}{A_S\left(\sigma_0^{NTC}\right)^2} = \rho_L\frac{\sigma_L}{\sigma_0^{NTC}} + \rho_S\frac{\sigma_S}{\sigma_0^{NTC}}. \tag{44}$$

Solving for the Sharpe ratio $\frac{\mu_0^{NTC}}{\sigma_0^{NTC}}$ and using Proposition 1, we have the results. ■

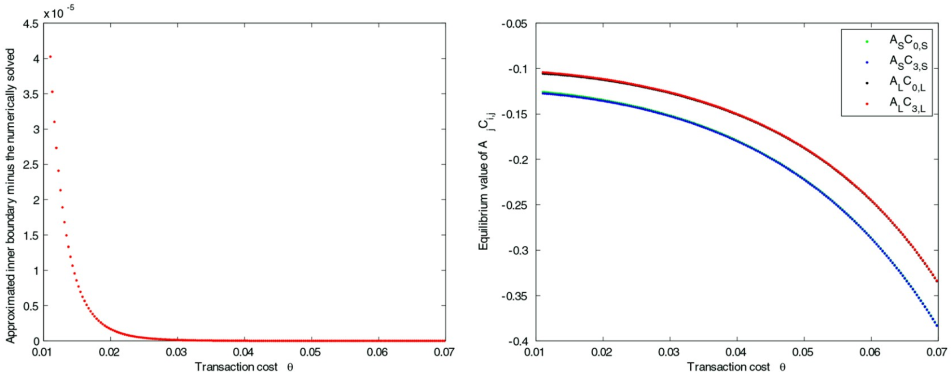
Interpretation: This frictionless equilibrium is a useful benchmark. It shows what risk-sharing looks like without costs, and it clarifies what transaction costs add.

9.2 Boundary approximation accuracy

Read: Figure A1 compares approximate inner trading boundaries with numerically solved boundaries.

- The approximation errors are small for the parameter values used in the main comparative statics.
- The figure also plots equilibrium values of terms involving marginal utility constants $A_iC_{j,i}$.
- This supports the use of Proposition 2 approximations in deriving Proposition 4.

Economic message: The analytical results are not purely formal; the approximation is numerically accurate in the relevant region.

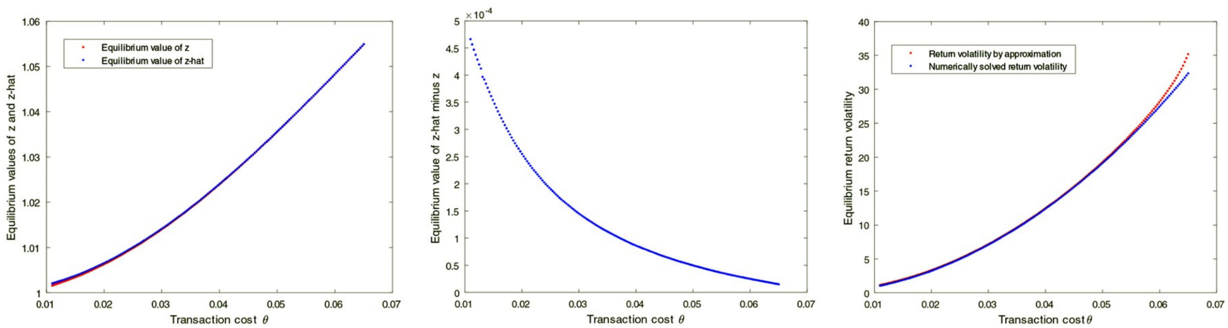


9.3 Endogenous volatility approximation

Read: Figure B1 compares the analytical equilibrium return volatility from Equation (31) with numerical equilibrium volatility.

- The left panel plots z and $\hat{z}$ as functions of transaction costs.
- The middle panel plots $\hat{z} - z$.
- The right panel compares analytical and numerical equilibrium volatility.
- When transaction costs are moderate, the analytical formula is close to the numerical solution.

Economic message: This validates the paper's central comparative statics: transaction costs raise the magnitude of equilibrium return volatility in the trading equilibrium.



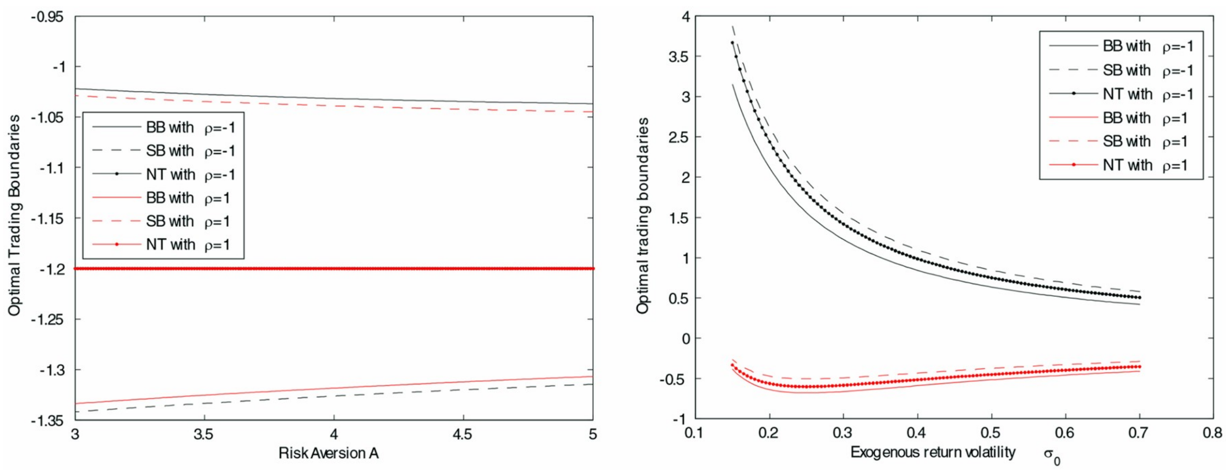
10 Figures and exhibits

10.1 Figure 1: individual trading boundaries

Read:

- The left panel plots buy and sell boundaries as functions of risk aversion.
- The right panel plots boundaries as functions of exogenous return volatility.
- The figure highlights asymmetry between long and short positions.
- It shows why matching long and short no-transaction regions is hard when agents have similar risk aversion.

Economic message: The micro-level boundary asymmetry is the foundation of the equilibrium results. It explains why transaction costs interact with risk-aversion heterogeneity.

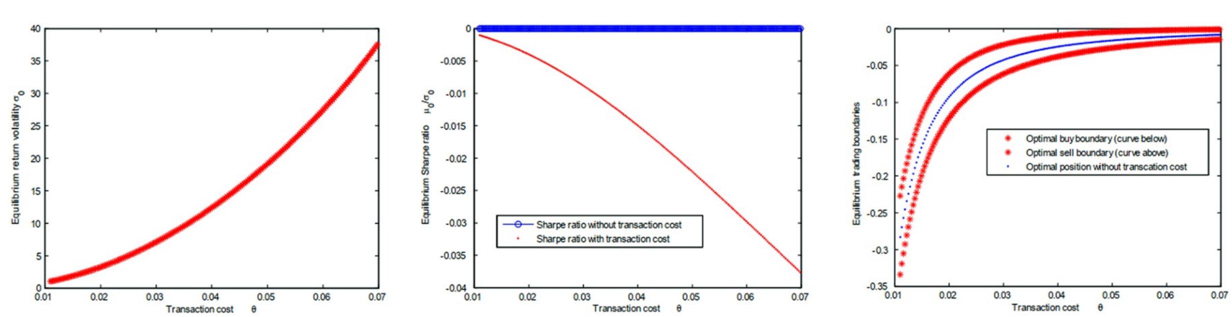


10.2 Figure 2: transaction costs and volatility

Read:

- Equilibrium return volatility increases with buy transaction costs.
- The Sharpe ratio and trading boundaries also adjust.
- The right panel shows no-transaction regions widening as transaction costs rise.

Economic message: This is the clearest visual statement of the main result: in a trading equilibrium, more transaction costs can mean more volatility.



10.3 Figures 3–4: risk-aversion heterogeneity

Read:

- Figure 3 shows how the equilibrium changes as one agent's risk aversion varies.
- Figure 4 decomposes transaction costs and expected utility changes.
- The figures show that trade is sustained when risk aversions are sufficiently different.
- When heterogeneity shrinks, transaction costs dominate hedging gains.

Economic message: Risk-sharing is not automatic in derivative markets with costs. Heterogeneity is needed for one side to supply liquidity to the other.

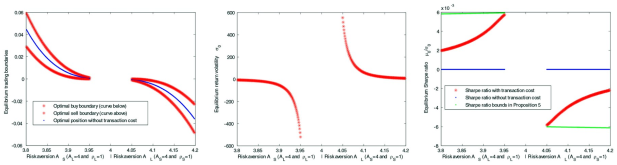


Figure 3.

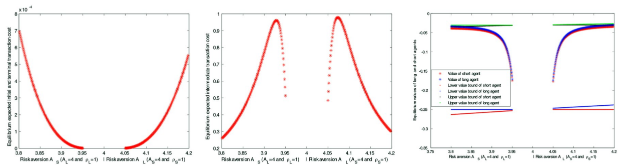


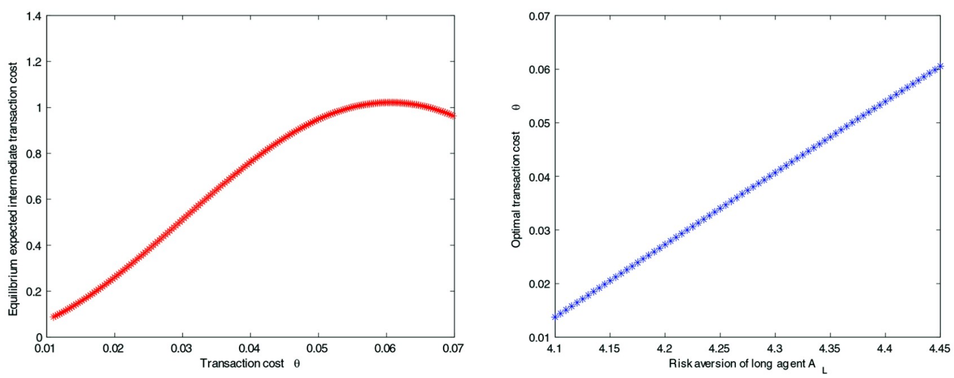
Figure 4.

10.4 Figure 5: revenue maximizing transaction cost

Read:

- Expected intermediate transaction cost is bell-shaped in the buy transaction cost.
- The revenue-maximizing transaction cost rises with risk-aversion heterogeneity.

Economic message: A revenue-maximizing transaction tax can exist at an interior point, but it may increase equilibrium volatility and reduce risk-sharing efficiency.



11 Short summary

- The paper builds an equilibrium model of derivative trading with proportional transaction costs.
- Agents trade a standardized zero-net-supply derivative to hedge nontraded endowment risk.
- Without transaction costs, each agent continuously holds a constant dollar position.
- With transaction costs, each agent uses a no-transaction region and trades only at buy/sell boundaries.

- Equilibrium with trade requires market clearing at both inner and outer boundaries.
- Transaction costs adjust the expected return differently at those boundaries.
- Equilibrium return volatility must adjust to reconcile the boundary-implied Sharpe ratios.
- When a trading equilibrium exists, the magnitude of return volatility increases in transaction costs.
- Trade with transaction costs requires sufficient heterogeneity in risk aversion.
- The less risk-averse agent typically takes the short side and earns a negative liquidity premium as compensation.
- Equilibria without trade always exist when transaction costs are positive.
- The model shows that financial transaction taxes can increase rather than decrease volatility in derivative markets.

12 Conclusion

12.1 Core conclusion

Loewenstein and Qin show that transaction costs in derivative markets can increase equilibrium return volatility. The result arises because equilibrium volatility is not fixed: it is the object that adjusts so that hedging demand, mean-variance demand, and market clearing can coexist in a market with no-transaction regions.

12.2 Economic intuition

Transaction costs prevent continuous rebalancing. Agents therefore tolerate imperfect hedging inside no-transaction bands. But the derivative is in zero net supply, so the long and short agents' trading boundaries must match. Transaction costs make the Sharpe ratio implied by one boundary differ from the Sharpe ratio implied by another boundary. Higher volatility changes both hedging and speculative demand, allowing the equilibrium to clear.

12.3 Takeaway

The paper cautions against a simple view that financial transaction taxes stabilize markets by reducing trading. In derivative markets where trading exists for risk-sharing reasons, higher transaction costs can make the contract more volatile and can eliminate trade altogether when heterogeneity is insufficient.